

ECO7101
Chapter 24
MONOPOLISTIC COMPETITION, OLIGOPOLY AND GAME THEORY

THE THEORY OF MONOPOLISTIC COMPETITION

Monopolistic Competition: A theory of market structure based on three assumptions: many sellers and buyers, firms producing and selling slightly differentiated products, and easy entry and exit.

The theory of monopolistic competition is built on three assumptions:

1. There are many sellers and buyers. This assumption holds for perfect competition too. For that reason, you might think that the monopolistic competitor should be a price taker, but it is a price searcher, basically because of the next assumption.
2. Each firm (in the industry) produces and sells a slightly differentiated product. Differences among the products may be due to brand names, packaging, location, credit terms connected with the sale of the product, the friendliness of the salespeople, and so forth. Product differentiation may be real or imagined. For example, aspirin may be aspirin, but if some people view a name brand aspirin (such as Bayer®) as better than a generic brand, product differentiation exists.
3. Entry and exit are easy. Monopolistic competition resembles perfect competition in this respect. There are no barriers to entry and exit, legal or otherwise. Examples of industries with monopolistic competition include retail clothing, computer software, restaurants, and service stations.

The Monopolistic Competitor's Demand Curve

The perfectly competitive firm has many rivals, all producing the same good, so the good that it produces has an endless number of substitutes. Accordingly, the elasticity of demand for its product is extremely high—so high, in fact, that the demand curve it faces is horizontal (for all practical purposes).

The monopoly firm has practically no rivals, and it produces a good that has no substitutes. The elasticity of demand for its product is low, as partly reflected by its downward-sloping demand curve.

The monopolistic competitor, like the perfectly competitive firm, has many rivals. But unlike the rivals of the perfectly competitive firm, the monopolistic competitor's rivals do not sell exactly the same product that the monopolistic competitor sells. Because its product has substitutes, but not perfect ones, the elasticity of demand for the monopolistic competitor's product is not as great as that of the perfectly competitive firm. Nor does its demand curve look like the one faced by the perfectly competitive firm. **The monopolistic competitor's demand curve is not horizontal; it is downward sloping.**

The Relationship between Price and Marginal Revenue for a Monopolistic Competitor

Because a monopolistic competitor faces a downward-sloping demand curve, it has to lower its price to sell an additional unit of the good it produces. (It is a price searcher. For example, let's say that it can sell 3 units at \$10 each but that it has to lower its price to \$9 to sell 4 units. Its marginal revenue is therefore \$6 (total revenue at 3 units is \$30, and total revenue at 4 units is \$36), which is below its price of \$9. **Thus, for the monopolistic competitor $P > MR$.**

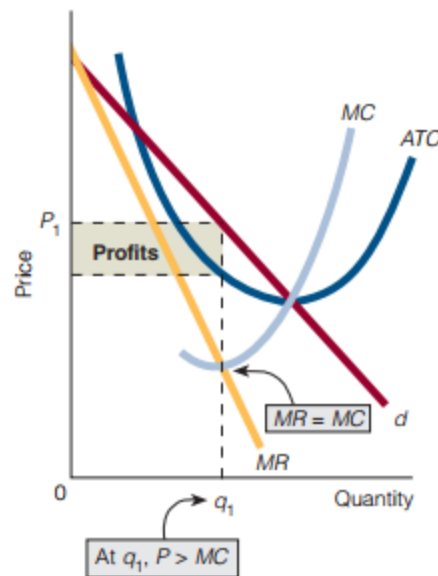
Output, Price, and Marginal Cost for the Monopolistic Competitor

The monopolistically competitive firm is the same as both the perfectly competitive firm and the monopoly firm in one regard: it produces the quantity of output at which $MR = MC$. In Exhibit 1, the firm produces 1 q . For this quantity, the monopolistic competitor charges the highest price it can charge: P_1 in the exhibit. For the monopolistic competitor, $P > MR$. Because the monopolistic competitor produces the quantity of output at which $MR = MC$, it must produce a level of output at which price is greater than marginal cost ($P > MC$), as is obvious in Exhibit 1.

EXHIBIT 1

The Monopolistically Competitive Firm's Output and Price

The monopolistic competitor produces that quantity of output for which $MR = MC$. This quantity is q_1 in the exhibit. The monopolistic competitor charges the highest price consistent with q_1 , namely, P_1 .



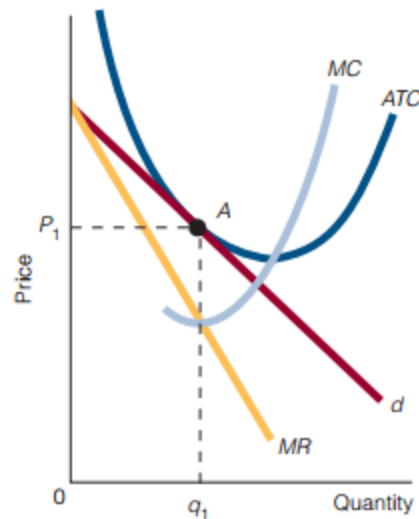
Will There Be Profits in the Long Run?

If the firms in a monopolistically competitive market are currently earning profits, as is the firm in Exhibit 1, most likely they will not continue to earn profits in the long run. The assumption of easy entry and exit precludes this possibility. If firms in the industry are earning profits, new firms will enter the industry and reduce the demand that each firm faces. In other words, the demand curve for each firm may shift to the left. Eventually, in the long run, competition will reduce economic profits to zero, as is shown for the monopolistically competitive firm in Exhibit 2.

EXHIBIT 2

Monopolistic Competition in the Long Run

Because of easy entry into the industry, there are likely to be zero economic profits in the long run for a monopolistic competitor. In other words, $P = ATC$.



Note, however, that the answer to the question of whether firms will continue to earn profits in the long run was “most likely” they won’t, instead of no. In monopolistic competition, new firms usually produce a close substitute for the product made by existing firms not an identical one. In some instances, this difference may be enough to upset the zero economic profit condition in the long run. An existing firm may be able to differentiate its product sufficiently in the minds of buyers such that it continues to earn profits, even though new firms enter the industry and compete with it.

Excess Capacity: What Is It, and Is It “Good” or “Bad”?

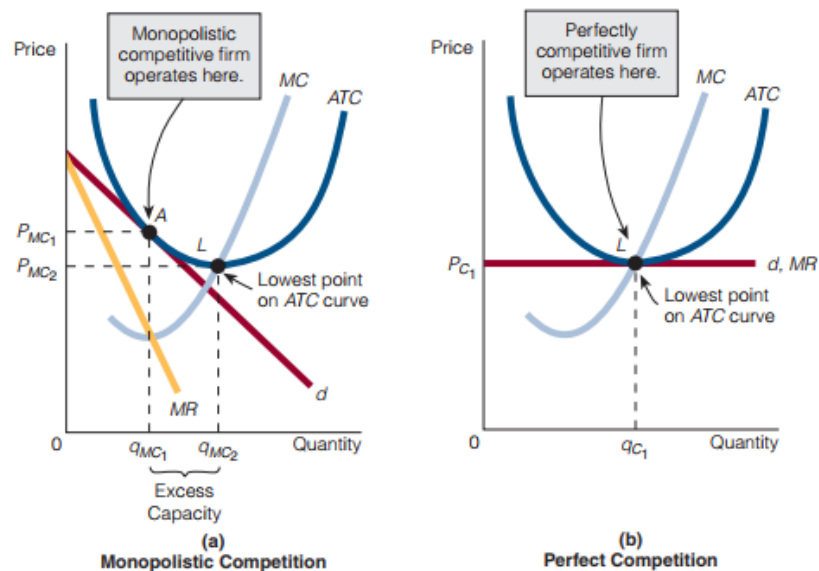
Excess Capacity Theorem: A monopolistic competitor in equilibrium produces an output smaller than the one that would minimize its costs of production.

The theory of monopolistic competition makes a major prediction, which is generally referred to as the excess capacity theorem: A monopolistic competitor will produce an output smaller than the one that would minimize its unit costs of production.

EXHIBIT 3

A Comparison of Perfect Competition and Monopolistic Competition: The Issue of Excess Capacity

The perfectly competitive firm produces a quantity of output consistent with lowest unit costs. The monopolistic competitor does not. If it did, it would produce q_{MC2} instead of q_{MC1} . The monopolistic competitor is said to underutilize its plant size or to have excess capacity.



At point A in Exhibit 3(a), the monopolistic competitor is in long-run equilibrium because profits are zero ($P = ATC$). Point A is not the lowest point on the average total cost curve; the lowest point is point L. Therefore, in long-run equilibrium, when the monopolistic competitor earns zero economic profits, it is not producing the quantity of output at which average total costs (unit costs) are minimized, given the scale of the plant. Exhibit 3 contrasts the perfectly competitive firm and the monopolistic competitor in long-run equilibrium. In part (b), the perfectly competitive firm is earning zero economic profits and price (P_C) equals average total cost (ATC). Furthermore, the point at which price equals average total cost (point L) is the lowest point on the ATC curve. In long-run equilibrium, the perfectly competitive firm produces the quantity of output at which unit costs are minimized.

Now look back at part (a). The monopolistic competitor is earning zero economic profits, and price (P_{MC1}) equals average total cost. If the monopolistic competitor produced the quantity of output at which unit costs were minimized, it would produce q_{MC2} . For this reason, it has been argued that the monopolistic competitor produces too little output (q_{MC1} instead of q_{MC2}) and charges too high a price (P_{MC1} instead of P_{MC2}). With respect to output, too little translates into the monopolistic competitor's underutilizing its present plant size; it is said to have excess capacity.

In part (a), the excess capacity is equal to the difference of q_{MC2} and q_{MC1} . Some have argued that the monopolistic competitor operates at excess capacity because it faces a downward-sloping demand curve. In Exhibit 3(a), the only way the firm would not operate at excess capacity is if its demand curve were tangent to the ATC curve at point L—the lowest point on the ATC curve. But for tangency at L to occur, the demand curve would have to be horizontal, a condition that would require homogeneous products. A downward-sloping demand curve cannot be tangent to the ATC curve at point L.

In short, the monopolistic competitor operates at excess capacity as a consequence of its downward-sloping demand curve, and its downward-sloping demand curve is a consequence of differentiated products.

Monopolistic competition is inefficient

Monopolistic competition is allocative inefficient and also it is productive inefficient. Monopolistic competition is allocative inefficient since it sets a price greater than price. it is productive inefficient because it does not produce capacity output in the long-run.

OLIGOPOLY MARKET

A theory of market structure based on three assumptions: few sellers and many buyers, firms producing either homogeneous or differentiated products, and significant barriers to entry.

The different theories of oligopoly have the following common assumptions:

1. There are few sellers and many buyers. The assumption is usually that the few firms of an oligopoly are interdependent; each one is aware that its actions influence the others and that the actions of the other firms affect it. This interdependence among firms is a key characteristic of oligopoly.
2. Firms produce and sell either homogeneous or differentiated products. Aluminum is a homogeneous product produced in an oligopolistic market; cars are a differentiated product produced in an oligopolistic market.
3. The barriers to entry are significant. Economies of scale constitute perhaps the most significant barrier to entry in an oligopoly, but patent rights, exclusive control of an

essential resource, and legal obstacles also act as barriers to entry. The oligopolist is a price searcher. Like all other firms, it produces the quantity of output at which $MR = MC$.

The Concentration Ratio

The percentage of industry sales (or assets, output, labor force, or some other factor) accounted for by x number of firms in the industry.

Which industries today are dominated by a small number of firms—that is, are oligopolistic?

Economists have developed the concentration ratio to help answer this question. The concentration ratio is the percentage of industry sales (or assets, output, labor force, or some other factor) accounted for by x number of firms in the industry. The number x in the definition is usually four or eight, but it can be any number (although it is usually small):

Four-firm concentration ratio: CR_4 = Percentage of industry sales accounted for by four largest firms

Eight-firm concentration ratio: CR_8 = Percentage of industry sales accounted for by eight largest firms

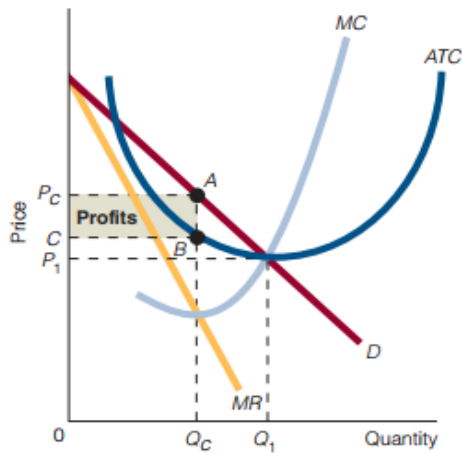
A high concentration ratio implies that few sellers make up the industry; a low concentration ratio implies that more than a few sellers make up the industry. As an example, let's calculate a four-firm concentration ratio for industry Z. Suppose total industry sales for a given year are \$5 million and the four largest firms in the industry account for \$4.5 million in sales. The four-firm concentration ratio is 0.90, or 90 percent (\$5million $\div$ \$4.5million = 0.90). Industries with high four- and eight-firm concentration ratios in recent years are cigarettes, cars, tires, cereal breakfast foods, farm machinery, and soap and other detergents, to name a few.

PRICE AND OUTPUT UNDER THE CARTEL THEORY

Cartel Theory: A theory of oligopoly in which oligopolistic firms act as if there were only one firm in the industry.

Cartel : An organization of firms that reduces output and increases price in an effort to increase joint profits.

The key behavioral assumption of the cartel theory is that, within a given industry, oligopolists increase price in an effort to increase joint profits.

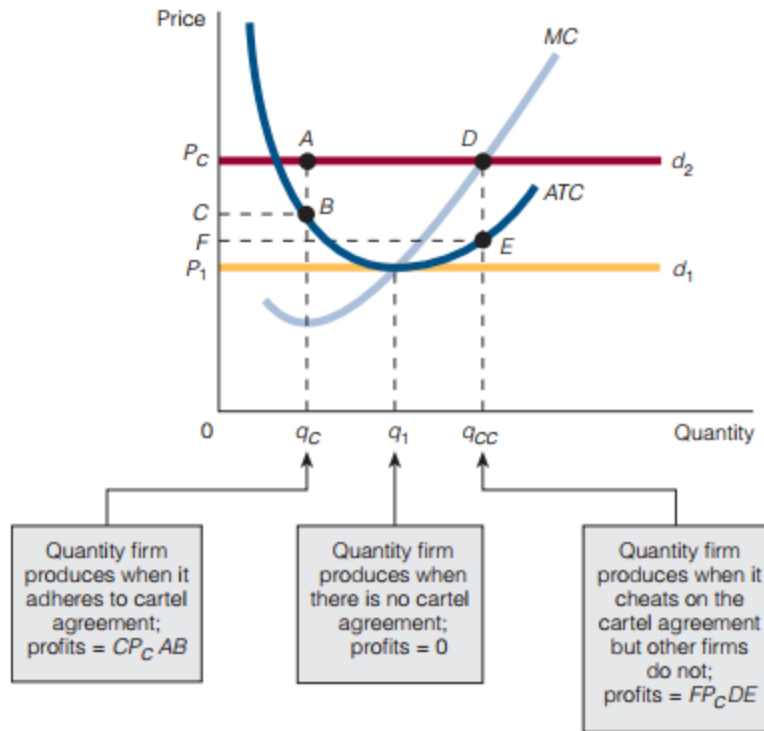


act as if there were only one firm. In short, they form a cartel to capture the benefits that would exist for a monopolist. A cartel is an organization of firms that reduces output and raises price.

Forming and maintaining a cartel has its benefits. Figure shows an industry in long-run competitive equilibrium where $MC = ATC = D$. Price is P_1 , and quantity of output is Q_1 . The industry is producing the output at which price equals marginal cost, and economic profits are zero. Now, suppose the firms making up the industry form a cartel and reduce output to Q_C . Then the new price is P_C (the cartel price), and profits, which are equal to the area CP_CAB , can be shared among the members of the cartel. With no cartel, there are no profits; with a cartel, profits are earned. Thus, the firms have an incentive to form a cartel and to behave cooperatively rather than competitively.

The Benefits of Cheating on the Cartel Agreement

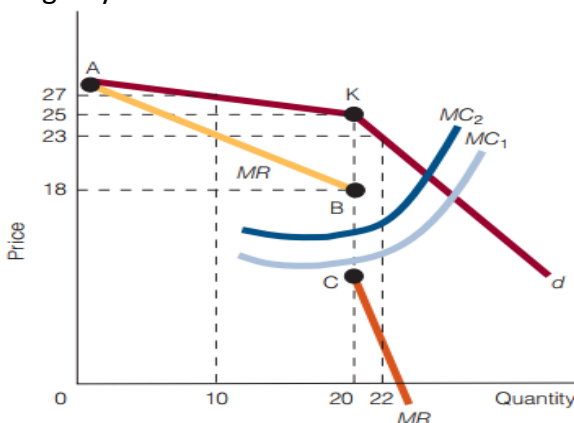
Firms may not be able to form a cartel, even though they have a profit incentive to do so. Also, even if they are able to form the cartel, the firms may not be able to maintain it. Firms that wish to form and maintain a cartel will encounter several problems.



In the exhibit we see the situation for a representative firm of a cartel. In long-run competitive equilibrium, the firm produces q_1 and charges P_1 , earning zero economic profits. As a consequence of the cartel agreement, it reduces its output to q_C and charges P_C . Its profits are the area CP_CAB . If a firm cheats on the cartel agreement and others do not, it will increase output to q_{CC} and reap profits of FP_CDE . Note, however, that if this firm can cheat on the cartel agreement, so can others. Given the monetary benefits gained by cheating, it is likely that the cartel will exist for only a short time.

The Kinked Demand Curve Theory

A theory of oligopoly that assumes, if a single firm in the industry cuts prices, other firms will do likewise, but if it raises price, other firms will not follow suit. The theory predicts price stickiness or rigidity.



The key behavioral assumption of the theory is that rival firms will not match a price hike but will match a price cut. The theory predicts that changes in marginal costs between B and C will not cause changes in price or output. It implies price rigidity.

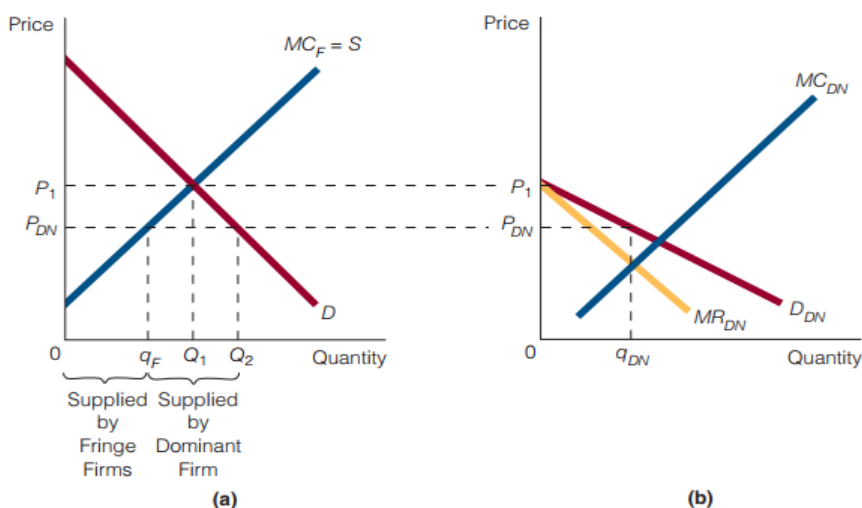
The current price charged by the firm is \$25. If the firm raises its price to \$27, other firms will not match it, and therefore the firm's sales will drop (from 20 to 10). In short, the demand curve for the firm above \$25 is highly elastic. However, if the firm lowers its price to, say, \$23, other firms will match the price cut, and therefore the firm's sales will not increase by much (only from 20 to 22). Demand is much less elastic below \$25 than above it. Thus there is a kink in the firm's demand curve at the current price (point K). The kink signifies that other firms respond radically differently to a single firm's price hikes than they do to its price cuts.

The kinked demand curve (and resulting MR curve) posits that prices in oligopoly will be less flexible (or more rigid) than in other market structures.

The Price Leadership Theory

In this theory of oligopoly, the dominant firm in the industry determines price, and all other firms take their price as given.

Suppose that there are ten firms in an industry, A–J, that firm A is the dominant firm, and that firm A is much larger than its rival firms. (The dominant firm need not be the largest firm in the industry; it could be the lowest-cost firm.) The dominant firm sets the price that maximizes its profits, and all other firms take this price as given. All other firms, then, are seen as price takers; thus, they equate price with their respective marginal costs. The price leadership explanation suggests that the dominant firm acts without regard to the other firms in the industry and simply forces the other firms to adapt. This assumption is not quite correct. The dominant firm sets the price based on information it has about the other firms in the industry, as shown in the figure.



Part (a) shows the market demand curve and the horizontal sum of the marginal cost curves of the fringe firms (all firms other than the dominant firm). Because these fringe firms are price takers, the marginal cost curve in (a) is their supply curve. The dominant firm observes that at a

price of P_1 , the fringe firms alone can supply the entire market. They will supply Q_1 . In short, P_1 and Q_1 define the situation in the industry or market that excludes the dominant firm.

Now add the dominant firm. It derives its demand curve, D_{DN} , by noting how much is left for it to supply at each given price. For example, at a price of P_1 , the fringe firms would supply the entire market, and nothing would be left for the dominant firm to supply. So a price of P_1 and an output of zero is one point on the dominant firm's demand curve, as shown in part (b). (Sometimes the dominant firm's demand curve is referred to as the residual demand curve for obvious reasons.) The dominant firm continues to locate other points on its demand curve by noting the difference between the market demand curve (D) and MC_F at each price below P_1 .

After the dominant firm calculates its residual demand curve, it produces the quantity of output at which its marginal revenue equals its marginal cost. This level is q_{DN} in (b). It charges the highest price for this quantity of output, which is P_{DN} . This is the price that the dominant firm sets and that the fringe firms take. Because they act as price takers, the fringe firms equate P_{DN} with marginal cost and produce q_F , as shown in part (a). The remainder of the total output produced by the industry—the difference between Q_2 and q_F —is produced by the dominant firm. Therefore, the distance from the origin to q_{DN} in (b) is equal to the difference between Q_2 and q_F in (a)